

SEC: Enabling MLLMs for Low-Latency IoT Video Analysis via Semantic-Aware Edge–Cloud Collaboration

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Abstract—The rapid proliferation of Internet of Things (IoT)-enabled cameras has driven increasing demand for low-latency, intelligent video understanding in real-world applications such as smart cities and industrial automation. While multimodal large language models (MLLMs) offer unprecedented capabilities in semantic reasoning and natural language-based video comprehension, their deployment in latency-sensitive IoT environments remains challenging due to high computational costs and sequential decoding bottlenecks. Moreover, conventional edge–cloud video analysis frameworks often rely on semantic-agnostic frame sampling, leading to information loss or redundant data transmission. In this article, we propose *SEC*, a semantic-aware edge–cloud collaborative framework for efficient and accurate video analysis. *SEC* introduces a task-aware key frame selection mechanism at the edge to maximize semantic relevance while minimizing bandwidth usage and a novel adaptive speculative decoding (SPD) framework with tree-based parallel generation on the cloud to accelerate MLLM inference. Extensive experiments under realistic edge–cloud deployment settings on four video understanding benchmarks demonstrate that the proposed *SEC* framework achieves superior performance, significantly reducing the end-to-end inference latency while improving the accuracy, a rare win-win in latency-critical IoT systems.

Index Terms—Frame selection, inference acceleration, speculative decoding (SPD), video analytics, vision-language model (VLM).

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I. INTRODUCTION

WITH the rapid advancement of Internet of Things (IoT) technologies, intelligent video perception has emerged as a cornerstone capability for critical applications such as smart cities, industrial automation, and surveillance systems [1], [2], [3], [4]. A massive number of camera-enabled IoT devices continuously generate video streams, driving the evolution of visual analysis from merely “seeing” to truly “understanding” scenes [5], [6], [7], [8]. Modern IoT deployments typically follow a hierarchical machine learning (ML) system architecture, where front-end sensors capture dense visual data, edge nodes conduct lightweight preprocessing, and cloud servers execute computation-intensive multimodal reasoning [9], [10], [11]. Within this IoT/ML pipeline, the timeliness of perception is critical; raw visual data must be transformed into high-level semantics quickly enough to support real-time decision-making, anomaly intervention, or automated control loops [5], [6], [7], [12], [13]. From raw visual input to high-level semantic understanding, the latency must be within strict constraints to meet the stringent timeliness requirements of real-world applications [6], [8], [14], [15].

Traditional video analysis approaches typically rely on handcrafted rules or task-specific models, such as object detection, action recognition, and optical flow analysis [3], [6], [15]. While these methods are effective in constrained settings, they often suffer from limited generalization and shallow semantic understanding, making them ill-suited for open and dynamic real-world environments [6], [7]. These methods are usually designed for isolated tasks, lacking the capacity to model complex natural language-based queries, and are unable to support interactive use [2], [15], [16]. As user demands for flexibility and general-purpose intelligence continue to grow, the limitations of conventional approaches, particularly in adaptability and cognitive expressiveness, have become increasingly apparent, rendering them inadequate for the evolving requirements of modern IoT systems.

Recently, the emergence of multimodal large language models (MLLMs) has brought transformative advances in video content understanding [17], [18], [19]. By deeply integrating visual encoders (e.g., ViT and CLIP) with large language models (LLMs) [20], [21], [22], MLLMs achieve cross-modal semantic alignment, enabling deep semantic parsing, temporal reasoning, and natural language question answering over video

content. For videos, MLLMs first encode spatiotemporal visual information through the visual encoder, which captures both frame-level features and motion patterns across time [23]. These visual representations are then projected into the semantic space of the language model, allowing the LLM to interpret and reason about the video content in a coherent, context-aware manner. In this manner, users can directly ask queries such as “Who opened the door and when?” or “What object appeared first in the video?” and the system can generate structured, human-readable responses, greatly enhancing the flexibility and intelligence of human-machine interaction.

However, due to the enormous parameter scale and computational overhead of MLLMs, deploying them directly on devices for end-to-end video analysis poses significant challenges in resource-constrained and latency-sensitive IoT scenarios [14], [15], [24]. Consequently, current mainstream approaches adopt an edge cloud collaborative architecture. Edge devices, such as cameras, edge gateways, and IoT sensors, capture raw video streams and perform lightweight preprocessing locally and then transmit critical clips/frames to the cloud, where powerful MLLMs conduct in-depth semantic understanding and complex reasoning [5], [6], [7], [12], [25]. This architecture effectively balances the resource limitations of edge devices with the computational capacity of the cloud and has become the dominant paradigm for large-scale intelligent visual services.

In this edge cloud video analysis pipeline, the clips/frames selection mechanism on the edge side plays a critical role [6]. The high frame rate of raw video streams, especially in multicamera, concurrent scenarios, leads to enormous data volume. Uploading all frames indiscriminately to the cloud would consume excessive bandwidth and significantly increase transmission latency and cloud-side processing load. To mitigate this, edge devices typically perform key frame selection or event-triggered sampling, transmitting only semantically relevant or visually salient video segments to reduce communication overhead and improve system scalability [3], [7], [12], [24], [26].

However, existing systems mostly rely on static or uniform frame sampling strategies, such as extracting frames at fixed time intervals [4], [7], [12], [24], [26]. While simple to implement and computationally efficient, these approaches fundamentally lack semantic awareness. They implicitly assume that meaningful content is uniformly distributed over time, whereas in reality, critical events, such as intrusions, falls, or switch operations, are often sudden, short-lived, and temporally localized [14], [27], [28]. For example, in a surveillance video lasting several minutes, an anomalous behavior may last only a few seconds, which uniform sampling can easily miss, leading to false negatives due to missing input at the cloud side. More critically, since the edge side sampling process is unaware of user queries or task objectives, the uploaded frames may be irrelevant to the actual inference task [7], [26]. This results in a dual waste: loss of critical information and transmission of redundant data, compromising both accuracy and resource efficiency. Such inefficiencies are further amplified in multicamera IoT deployments, where dozens or even hundreds of streams must share limited uplink

bandwidth, making semantic-aware edge-side filtering crucial for system scalability.

In addition, the cloud-deployed MLLMs suffer from significant inference latency, primarily caused by the inherent decoding constraints of LLMs [5], [6], [7], [12]. The sampled frames are uploaded to the cloud, encoded by a vision encoder, and concatenated with a text query for joint reasoning in the LLM. Existing LLMs, predominantly based on the Transformer architecture [29], rely on auto-regressive decoding as their standard generation paradigm. During decoding, each generated token depends on all previously produced tokens, requiring the LLM to perform multiple forward passes in a strictly sequential manner, generating only one token per forward pass [30], [31], [32]. Unlike the prefill phase, where input tokens are processed in parallel, this decoding process is inherently sequential and cannot be parallelized. For tasks demanding multihop reasoning, detailed descriptions, or complex logical inference, the entire output sequence must be generated iteratively, resulting in latency that scales linearly with output length. In cloud-hosted IoT services, where numerous edge devices simultaneously issue queries, this decoding bottleneck directly limits overall throughput and response time, becoming one of the major system-level constraints.

In this article, we propose a semantic-aware edge-cloud collaborative framework for low-latency IoT video analysis services, named *SEC*. To minimize end-to-end latency, our framework optimizes both the video preprocessing and the decoding pipeline at the edge and the cloud. Specifically, on the edge side, we introduce a semantic-aware frame selection module that identifies task-relevant key frames while filtering out redundant or irrelevant ones. *SEC* significantly improves the information density and semantic relevance of the uploaded frames. On the cloud side, we design an adaptive speculative decoding (SPD) framework for MLLMs. It leverages generation confidence and visual priors to dynamically decide when to verify candidate tokens, enabling more efficient use of computation. Furthermore, we introduce a tree-based parallel SPD mechanism that generates diverse candidate sequences in parallel, to provide more candidates for improving speculative acceptance, leading to a substantial boost in inference throughput.

Extensive experiments under realistic edge-cloud deployment settings on four video understanding benchmarks, VideoMME [39], LongVideoBench [40], VideoMME-CP [39], and Perception Test [41], demonstrate that the proposed *SEC* framework achieves superior performance, significantly reducing the end-to-end inference latency while improving the accuracy, a rare win-win in latency-critical systems. Notably, *SEC* is a training-free framework that requires no fine-tuning of the underlying MLLMs, making it highly efficient to deploy and easily transferable to arbitrary MLLMs. This plug-and-play compatibility significantly enhances its practicality for real-world IoT applications.

The main contributions of our work are as follows.

- 1) We propose a semantic-aware edge-cloud collaborative framework, with adaptive edge frame selection and cloud decoding, for low-latency IoT video analysis services, named *SEC*.

TABLE I
SUMMARY OF KEY ASPECTS OF EXISTING METHODS WHEN EXTENDED TO EDGE-CLOUD VIDEO ANALYSIS

References	Sampling Type	Decoding Type	Edge-friendly	Query-aware	Coverage-aware	Tree-based	Training-free
[33]	Static	Auto-regressive	✓	×	✓	-	✓
[34]	Adaptive	Auto-regressive	×	×	×	×	✓
[14]	Adaptive	Auto-regressive	×	✓	✓	×	×
[28]	Adaptive	Auto-regressive	✓	✓	×	×	×
[35]	Adaptive	Auto-regressive	×	✓	×	×	×
[36]	Adaptive	Auto-regressive	×	✓	×	×	✓
[37]	Adaptive	Auto-regressive	×	✓	✓	×	✓
[30]	Static	Speculative decoding	✓	×	×	×	✓
[38]	Static	Speculative decoding	✓	×	✓	✓	×
[32]	Static	Speculative decoding	✓	×	✓	✓	✓
Our <i>SEC</i>	Adaptive	Speculative decoding	✓	✓	✓	✓	✓

- 2) On the edge, we design a lightweight semantic-aware frame selection module that leverages visual saliency and query-relevant cues to identify and transmit only task-critical frames, effectively filtering out redundant or irrelevant content while preserving essential spatiotemporal information for downstream reasoning.
- 3) On the cloud side, we introduce a novel inference acceleration framework that integrates adaptive SPD with a tree-based parallel generation mechanism, enabling concurrent exploration of diverse candidate token sequences and significantly improving the token acceptance rate.
- 4) Extensive experiments under realistic edge-cloud deployment settings show that our training-free *SEC* framework significantly reduces inference latency while improving accuracy, and seamlessly generalizes to arbitrary MLLMs under realistic edge-cloud deployments.

II. RELATED WORK

A. Large-Scale Vision-Language Models

In contrast to models trained solely with image supervision, vision-language models (VLMs) [42], [43], [44], [45], [46] effectively synthesize information from diverse modalities, driving significant progress in cross-modal tasks of vision and language. Early works primarily concentrate on vision-language learning, exemplified by models such as ViT [42], CLIP [43], ALIGN [47], FILIP [48], and Florence [49], demonstrating exceptional performance on a wide spectrum of tasks, including few-shot and zero-shot visual recognition. Subsequently, inspired by the remarkable capabilities of LLMs, a series of studies adopt pretrained LLMs [20], [21], [50], [51], [52], [53] for multimodal processing by splitting and fine-tuning specific components of LLMs [54], [55], such as layers, attention heads, or feed-forward networks. Another line explores extending pretrained LLMs into VLMs by incorporating additional vision encoders [45], [46], [56], which leverage MLP projectors or Q-formers to bridge the vision encoder with the LLM. More recently, researchers have begun to explore the design of architectures that eliminate the need for separate visual encoders to achieve unified multimodal modeling [57], [58], [59]. As research progresses, increasing efforts are being directed toward making these models more practical for real-world deployment. However, despite their impressive generalization and reasoning capabilities, most existing VLMs are designed under the assumption

of abundant computation and memory on the cloud side. In real-world IoT deployments, edge devices operate under tight bandwidth budgets and limited compute capacity, making the direct application of these models infeasible. This motivates the exploration of architectures and algorithms that can cooperate across the edge-cloud hierarchy to deliver real-time multimodal understanding. In this article, we explore how to accelerate VLMs while maintaining decoding accuracy on the cloud.

B. Speculative Decoding

In autoregressive models, the iterative, step-by-step generation process can be computationally expensive and time-consuming, particularly for large-scale models. To parallelize this inherently sequential decoding process, SPD was initially proposed in the form of block-wise parallel decoding [60]. Further advancements have introduced methods based on independent draft models as well as draft-free approaches, exploring the potential of SPD [30], [31], [61], [62], [63]. Additionally, some studies have explored the use of tree-based parallel verification of multiple candidate sequences [32], [38], [64], [65], [66]. Regarding the verification positions, existing methods predominantly adopt fixed-size verification windows. While simple, this fixed-size verification is content-agnostic and lacks adaptability. Some studies leverage statistical information, such as confidence scores and acceptance rates of draft tokens [62], [65], [67], to dynamically adjust the length of draft sequences. While effective in certain scenarios, these methods remain constrained by their reliance on textual information alone, leaving room for further improvement. Moreover, most SPD methods have been developed primarily for text-only LLMs. They overlook the characteristics of multimodal reasoning, where visual inputs influence semantic grounding and attention dynamics during token generation. In IoT video analysis scenarios, where cloud-side MLLMs must serve numerous concurrent queries from distributed cameras, relying solely on textual heuristics becomes insufficient. This highlights the need for SPD strategies that incorporate multimodal priors and can operate efficiently under system-level throughput constraints. In this article, we explore how to achieve content-aware adaptive verification in VLMs, utilizing not only textual statistics but also visual priors, for efficient IoT service.

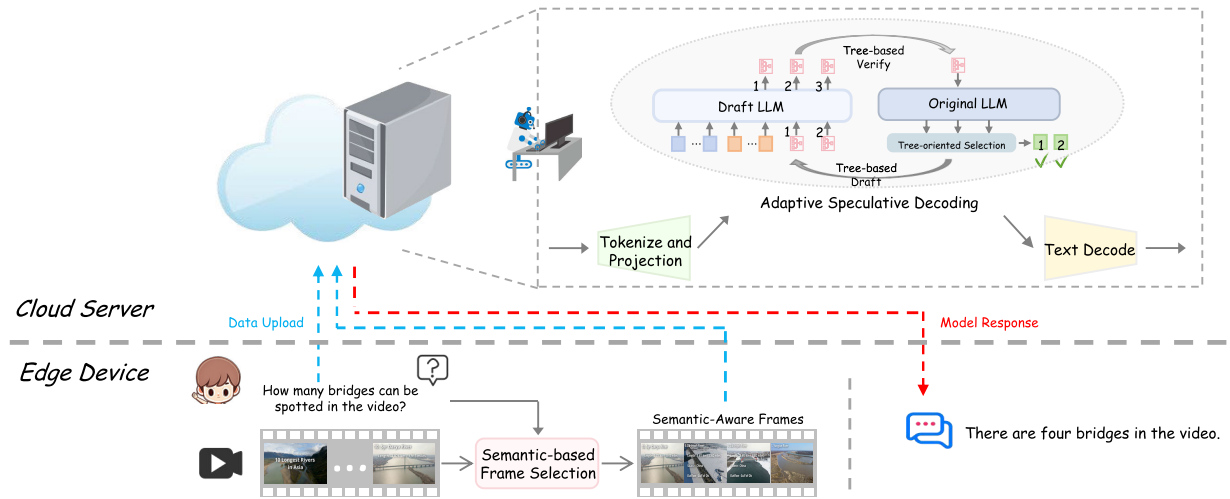


Fig. 1. Overview of our edge-cloud collaborative framework *SEC*. Our framework enables fast and accurate multimodal reasoning by collaborative computation between edge and cloud: 1) on the edge, a semantic-aware frame selection module filters irrelevant frames based on the user query, transmitting only keyframes to the cloud and 2) in the cloud, an adaptive SPD accelerates MLLM inference via tree-based parallel draft generation and verification, reducing latency while preserving accuracy.

C. Video Frame Selection

Video frame selection has been a long-standing research topic and has attracted considerable attention in the field of video understanding [14], [27], [68], [69], [70], well before the emergence of MLLMs. Early approaches relied on reinforcement learning [14], [27], [28], [68], where an agent previews the video and selects key frames based on learned policies. These methods are typically nonsemantic in nature, with the optimization objective being to maximize the probability of the correct label, thus lacking generalization ability for open-vocabulary tasks. Subsequently, several end-to-end trainable methods were proposed [69], [70], [71], leveraging techniques such as Gumbel-Softmax and bilinear interpolation to enable differentiable frame selection. More recently, the advent of CLIP [43] has provided a powerful tool for open-vocabulary image-text matching. Inspired by this, various CLIP-based frame selection strategies have been explored [72], [73], yielding promising results.

With the rapid development of image MLLMs, increasing efforts have been devoted to designing video MLLMs that can better handle temporal content [18], [19], [23], [73], [74], [75], [76], [77], [78]. Most existing video MLLMs adopt uniform sampling, extracting sparse frames at fixed intervals from dense video sequences. Only a few works have explored more adaptive frame selection strategies [35], [36], [37], [73]. Most prior frame selection approaches are developed for offline video understanding or cloud-only inference, where computational resources are abundant and bandwidth constraints are negligible. However, in IoT deployments, edge devices must process continuous long-duration video streams under limited power and communication budgets. This system-level constraint necessitates frame selection strategies that are lightweight, training-free, and capable of operating directly on heterogeneous edge hardware while preserving semantic alignment with downstream MLLMs. As shown in Table I, in this work, we focus on building a training-free and edge-friendly frame selection strategy that can efficiently improve

the performance of MLLMs on long-form videos, enabling efficient real-world IoT analytics pipelines.

III. OVERVIEW

To achieve low-latency video understanding in IoT systems, we propose a semantic-aware edge-cloud collaborative framework, named *SEC*, that jointly optimizes both the edge-side preprocessing and cloud-side inference of MLLMs. Modern IoT deployments typically follow a hierarchical system pipeline such that front-end cameras continuously stream raw visual data, edge nodes conduct lightweight filtering or compression, and cloud servers execute computation-intensive multimodal reasoning. Within this pipeline, reducing communication overhead and shortening the sequential decoding delay of MLLMs are both crucial for meeting real-time IoT systems.

As illustrated in Fig. 1, given an input video $\mathcal{V}$ and a natural language query $\mathcal{Q}$, the semantic-aware frame selection module on the edge server first efficiently selects a compact set of representative frames from a long video. The selected frames are both semantically relevant to the given question and temporally diverse across the entire video duration. The module is training-free and plug-and-play, making it flexible to develop on edge. Then, the query-aligned frames are uploaded to the cloud.

On the cloud side, the selected frames $\mathcal{F}$ and query $\mathcal{Q}$ are fed into an MLLM for joint reasoning. To accelerate the auto-regressive generation process, we introduce an adaptive SPD method. The draft sequences generated by the draft model $\mathcal{M}_d$ are organized into a tree-structured topology that enables parallel generation of diverse candidate sequences. This richer set of candidates increases the likelihood of speculative acceptance, leading to a substantial improvement in inference throughput. Moreover, unlike conventional SPD methods that use a fixed verification interval, we propose a novel content-aware routing module that dynamically determines when to invoke the powerful target model $\mathcal{M}_o$ for

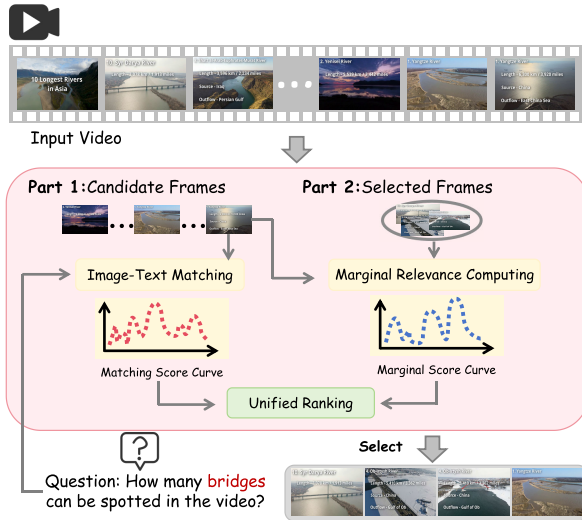


Fig. 2. Semantic-based edge frame selection. We propose a query-driven frame selection mechanism that operates entirely on the edge, ensuring that only the most informative frames are uploaded. Our method unifies and ranks the frame-query matching score and the marginal relevance score to balance semantic relevance and temporal diversity during selection.

validation. By monitoring generation confidence and shifts in visual attention, this module adapts the verification timing to the semantic complexity of the ongoing inference, striking a better balance between efficiency and output fidelity. With semantic-aware frame selection at the edge and adaptive tree-based SPD in the cloud, our framework significantly reduces end-to-end latency while preserving high semantic fidelity and reasoning accuracy, being well-suited for real-time IoT video analysis.

IV. EDGE FRAME SELECTION OPTIMIZATION

In IoT deployments, raw video streams captured by distributed cameras often contain high frame rates and long durations, yet only a small portion of frames are truly relevant to a given query. Uploading all frames not only saturates the limited uplink bandwidth but also overloads the visual encoder and MLLM with redundant content, degrading both latency and reasoning accuracy.

To overcome this, as shown in Fig. 2, we design a query-driven frame selection mechanism that operates entirely on the edge device. As shown in Algorithm 1, this mechanism ensures that only the most informative and query-relevant frames are uploaded, significantly reducing transmission cost in resource-constrained IoT scenarios.

Given a video $\mathcal{V}$ of T frames, the edge mechanism first splits the whole video into M nonoverlapping temporal segments. From each segment, the mechanism selects the central frame to form the candidate frame set $\bar{\mathcal{F}} = \{f_1, f_2, \dots, f_M\}$, where each frame $f_i \in \mathbb{R}^{H \times W \times C}$. This ensures that the candidate set spans the entire video sequence, providing broad temporal coverage.

A. Frame-Query Matching

The first step toward semantic awareness is measuring how well each candidate frame aligns with the user's intent, as expressed in the natural language query $\mathcal{Q}$. Given a question

Algorithm 1 Semantic-Aware Edge Frame Selection

Input: Video stream V , Query $\mathcal{Q}$, Segment number M , Selection number N , Factors α, γ

Output: Selected Key Frames $\mathcal{F}$

Divide V into M segments and extract central frames as candidate set $\bar{\mathcal{F}} = \{f_1, \dots, f_M\}$;

Initialize selected set $\mathcal{F} \leftarrow \emptyset$;

Extract visual embeddings $\{v_i\}$ for $\bar{\mathcal{F}}$ and text embedding t_q for $\mathcal{Q}$;

Compute semantic matching scores $\text{SIM}_i \leftarrow \text{Cosine}(v_i, t_q)$ for all $i \in [1, M]$;

while $|\mathcal{F}| < N$ **do**

for candidates $\bar{f}_i \in \bar{\mathcal{F}} \setminus \mathcal{F}$ **parallel do**

if $\mathcal{F} = \emptyset$ **then**

$\text{MRS}_i \leftarrow 0$;

else

 Compute Temporal Distance $D_t(\bar{f}_i, \mathcal{F})$ as Eq.4;

 Compute Visual Uniqueness $D_v(\bar{f}_i, \mathcal{F})$ as Eq.5;

 Compute Marginal Relevance

$\text{MRS}_i \leftarrow \alpha \cdot D_t + (1 - \alpha) \cdot D_v$;

 Calculate Unified Score

$S_i^u \leftarrow \gamma \cdot \text{SIM}_i + (1 - \gamma) \cdot \text{MRS}_i$;

 Select candidate frame f^* with maximum S^u ;

 Update $\mathcal{F} \leftarrow \mathcal{F} \cup \{f^*\}$;

return $\mathcal{F}$;

$\mathcal{Q}$ and candidate frame $\bar{f}_i$, we leverage pretrained VLMs (e.g., CLIP, TinyCLIP, or SigLIP) to project frames and text into a shared embedding space

$$t_q = E_t(\mathcal{Q}), v_i = E_v(\bar{f}_i). \quad (1)$$

The semantic similarity between the question $\mathcal{Q}$ and the i th candidate frame is measured via cosine similarity

$$\text{SIM}_i = \text{cosine} \left(\frac{t_q}{\|t_q\|}, \frac{v_i}{\|v_i\|} \right). \quad (2)$$

These scores $S_{\text{match}} = \text{SIM}_0, \text{SIM}_1, \dots, \text{SIM}_M$ provide a preliminary ranking of frames by their relevance to $\mathcal{Q}$, forming the foundation for intelligent selection.

B. Marginal Relevance Computing

While the above frame-query matching provides a strong indication of semantic relevance, selecting frames solely based on S_{match} may result in a set of frames that are clustered in time and visually similar. This greedy matching strategy can cause the cloud-side MLLM to miss important contextual dynamics, limiting comprehensive scene understanding.

Therefore, we introduce marginal relevance, aiming to balance semantic relevance and temporal diversity during selection. Given unselected frame f_i and the selected set $\mathcal{F} = \{f_1, \dots, f_K\}$, its marginal relevance is defined as

$$\text{MRS}_i = \alpha \cdot D_t(\bar{f}_i, \mathcal{F}) + (1 - \alpha) \cdot D_v(\bar{f}_i, \mathcal{F}) \quad (3)$$

where $D_t(\cdot)$ and $D_v(\cdot)$ denote the temporal distance and visual uniqueness. Tradeoff factor $\alpha \in [0, 1]$ controls the tradeoff between temporal and visual diversity.

Temporal distance $D_t(\cdot)$ measures how far $\bar{f}_i$ is from previously selected frames in time. To keep the comparable range, we normalize the distance as

$$D_t(\bar{f}_i, \mathcal{F}) = \frac{\min_{f_j \in \mathcal{F}} |I(\bar{f}_i) - I(f_j)|}{M} \quad (4)$$

where $I(f_i)$ denotes the absolute index in video sequence of frame f_i and M denotes the number of frames in the video. The temporal normalization ensures that the distances are comparable across videos of different lengths, avoiding bias toward longer videos. A larger value indicates greater temporal separation, improving coverage.

Visual uniqueness $d_v(\cdot)$ evaluates how distinct $\bar{f}_i$ is from existing selections

$$D_v(\bar{f}_i, \mathcal{F}) = 1 - \max_{j \in \mathcal{F}} (\text{cosine}(v_i, v_j)). \quad (5)$$

The maximum similarity is taken to reflect the most similar selected frame. This prevents the selection of visually similar frames, ensuring richer context for downstream reasoning.

C. Unified Ranking

With both semantic relevance and marginal diversity quantified, we now integrate them into a unified selection strategy. Rather than selecting all frames at once, we adopt an iterative greedy approach that dynamically updates the selected set $\mathcal{F}$, allowing each decision to benefit from prior context. This progressive selection process is lightweight and well-suited for edge devices, while ensuring that the uploaded subset provides sufficient context for comprehensive understanding in the cloud.

At iteration n , based on the frame-query matching score SIM_i and the marginal relevance score MRS_i , the unified score of candidate frames i from $\bar{\mathcal{F}}$ is defined as

$$S_i^u = \gamma \cdot \text{SIM}_i + (1 - \gamma) \cdot \text{MRS}_i \quad (6)$$

where $\gamma \in [0, 1]$ balances query alignment and diversity. The candidate with the highest S_i^u is added to $\mathcal{F}$, and MRS_i is recomputed for the remaining candidates to reflect the updated selection state. This iterative process continues until N key frames are selected. The resulting subset $\mathcal{F} = \{f_1, \dots, f_N\}$ is not only semantically aligned with the query Q but also temporally dispersed and visually diverse, providing a compact yet comprehensive visual context for cloud-side reasoning.

Once $\mathcal{F}$ is finalized, the selected frames are encoded and transmitted to the cloud, where they are fused with the user query Q and fed into the MLLM for multimodal reasoning.

D. Multiturn Frame Selection and Incremental Update

In real-world MLLM applications, user interactions frequently unfold across multiple conversational turns. After answering the initial query $Q^{(1)}$, the cloud MLLM may receive follow-up queries, each potentially shifting the semantic focus. To support such multiturn scenarios efficiently, our edge-side selection module performs incremental frame uploading by reusing previously uploaded frames.

Let $\mathcal{F}^{(p)}$ denote all past frames transmitted in earlier turns. For the current-turn selection set $\mathcal{F}^{(h)}$, the edge uploads only

the newly selected and previously unseen frames $\mathcal{U}^{(h)} = \mathcal{F}^{(h)} \setminus \mathcal{F}^{(p)}$. Thus, the cumulative transmission cost becomes $\text{Upload}^{(H)} = \bigcup_{h=1}^H \mathcal{U}^{(h)}$, which is substantially smaller than retransmitting a fresh selection at every turn.

When user queries rapidly across turns, our incremental update mechanism remains stable under this scenario. If the semantic overlap is high, frame reuse dominates and transmission cost is minimized. If the semantic overlap is low, the system gracefully degrades to uploading a fresh yet compact set of query-relevant frames. Since the per-turn selection size $|\mathcal{F}^{(h)}|$ is strictly bounded, the worst case communication overhead per turn remains controlled, preventing unbounded bandwidth growth even under rapidly changing user intents.

E. Analysis of Information Utility

In high-frame-rate IoT streams, visual content changes slowly between adjacent frames, implying high temporal redundancy. For a specific query Q , the information utility of our selected subset $\mathcal{F}$ is expected to approximate that of the full video $\mathcal{V}$ due to the inherent semantic redundancy of video streams. Uniform sampling reduces data size but blindly discards frames regardless of their content, potentially missing short-lived but critical events (the “needle in the haystack” problem).

In contrast, our semantic-aware selection acts as a relevance filter. By maximizing the joint objective of semantic similarity and marginal relevance as (6), our method indicates that: 1) *relevance guarantee*—every selected frame $f_i \in \mathcal{F}$ satisfies a minimum relevance score to Q , ensuring that no unrelated noise consumes bandwidth; and 2) *coverage guarantee*—the inclusion of marginal relevance forces the selected frames to span the semantic space of $\mathcal{V}$.

Therefore, although $|\mathcal{F}| \ll |\mathcal{V}|$, this analysis offers an intuition for why the selected frames retain most of the inferential cues needed by the downstream MLLM, which is empirically validated in Sections VI-B and VI-E through comparable accuracy with significantly reduced bandwidth.

V. CLOUD MLLM SPECULATIVE ACCELERATION

Even with optimized visual input from the edge, cloud-side MLLM inference remains a major bottleneck due to the inherently sequential nature of autoregressive decoding. In IoT video analytics, where numerous edge devices continuously issue queries, the decoding throughput of the cloud LLM becomes a central factor determining system scalability and response latency.

To break this sequential dependency, as shown in Fig. 3, we propose an adaptive SPD framework that enables low-latency token generation while maintaining high acceptance accuracy. Our method fits naturally into the cloud tier of the IoT/ML pipeline, enabling substantially higher throughput under concurrent multistream workloads.

A. Window-Free Adaptive Routing

Traditional SPD typically relies on fixed verification intervals (e.g., every five tokens), regardless of the input content or generation dynamics. This one-size-fits-all approach is

inherently inefficient, which may trigger verification too early during stable, predictable output sequences or too late after a semantic shift has already occurred in the multimodal context. It can waste potential speculation gains, leading to cascading errors and low acceptance rates.

To address this, we introduce a content-aware routing module M_r that operates on the cloud side and dynamically determines whether to continue drafting or initiate verification. Unlike prior methods that treat speculation as a language-only process, our module explicitly accounts for the multimodal context formed by the edge-selected video frames and the user query $\mathcal{Q}$. By adapting to both linguistic and visual signals, M_r enables timely and accurate speculation control, which is critical for low-latency MLLM services in IoT video analysis.

At the draft step l , given the draft logits P_l^{logit} and normalized visual attention distribution A_l^{vis} over visual tokens, with $l \in [1, k]$, the content-aware routing module M_r evaluates the draft tokens from two aspects, i.e., confidence enough? and Attention shift?.

1) *Confidence Evaluation*: Low-confidence predictions suggest uncertainty and a higher risk of mismatch. Specifically, the routing module M_r measure confidence via the entropy of draft logits P_l^{logit} as

$$H(P_l^{\text{logit}}) = - \sum_j p_j \log(p_j) \quad (7)$$

where p_j represents the predicted probability of the j th token of P_l^{logit} .

Then, a weighted entropy combination is used to merge the entropies of all k tokens of the current drafting

$$H_{\text{all}} = \beta \cdot H_{\text{avg}} + (1 - \beta) \cdot H_{\text{max}} \quad (8)$$

where H_{avg} and H_{max} denote the mean and maximum entropy across all k tokens, respectively, and $\beta \in [0, 1]$ is a tradeoff factor.

Lower entropy indicates higher confidence, reflecting a more peaked and decisive probability distribution over the token predictions. Conversely, higher entropy suggests uncertainty, signaling the need for verification by the original model. To filter out less confident candidates while accommodating the varying stages of the model's generation process, we introduce an adaptive temperature annealing and warming threshold τ_d , formalized as follows:

$$R_d = \begin{cases} 1, & \text{if } H_{\text{all}} \geq \tau_d \\ 0, & \text{if } H_{\text{all}} < \tau_d \end{cases} \quad (9)$$

where 1 indicates routing to *Verify Tokens* and 0 indicates and *Draft More*. The threshold τ_d starts from a low initial value τ_d^{initial} to probe the model's generation distribution. We define a learning rate lr to support the adaptive adjustment of τ_d . At each drafting step i , given the entropy H_{all} , the τ_d is updated as

$$\tau_d = \min(\hat{\tau}_d, \tau_d + lr \cdot (H_{\text{all}} - \tau_d)). \quad (10)$$

τ_d evolves dynamically until it reaches its upper limit $\hat{\tau}_d$, ensuring it adapts to the model's confidence levels during

decoding. Furthermore, it also unifies the confidence evaluation for both short-sequence and long-sequence generation tasks.

2) *Attention Shift Tracking*: The edge-selected frames $\mathcal{F}$ constitute a compact yet semantically aligned visual context. When the model's visual grounding changes substantially, either across spatial regions or across video frames, the previously drafted tokens may become unreliable or obsolete for continuation. The routing module M_r uses the centroid of the visual attention distribution over visual tokens A_l^{vis} to evaluate whether an attention shift has occurred between the current and previous iterations.

Modern MLLMs generate visual tokens from heterogeneous sources, including patch embeddings [18], Q-former queries [79], or multiscale pyramid features [80]. To obtain a consistent spatial reference, we project all visual tokens of each frame onto a canonical 2-D grid of resolution $H_c \times W_c$, regardless of their original feature-map size. $H_c \times W_c$ denotes the maximum spatial resolution among all visual feature levels used for token generation. Specifically, for a visual token j originating from a coarser feature level, $H' \times W'$, with spatial coordinate (u_j, v_j) , its canonical coordinates $(\tilde{u}_j, \tilde{v}_j)$ are computed as

$$\tilde{u}_j = \left\lfloor \frac{u_j}{H'} H_c \right\rfloor, \quad \tilde{v}_j = \left\lfloor \frac{v_j}{W'} W_c \right\rfloor. \quad (11)$$

This normalization unifies multiscale visual layouts and enables model-agnostic spatial attention analysis.

To further incorporate temporal structure, we augment each token with a temporal coordinate t_j representing its frame index. Thus, each token is associated with a 3-D position, $(\tilde{u}_j, \tilde{v}_j, t_j)$, forming a canonical spatio-temporal grid. To ensure a consistent scale across the spatial and temporal axes, we further normalize the three coordinates into a unified $[0, 1]$ range, i.e.,

$$\hat{u}_j = \frac{\tilde{u}_j}{H_c}, \quad \hat{v}_j = \frac{\tilde{v}_j}{W_c}, \quad \hat{t}_j = \frac{t_j}{M} \quad (12)$$

where M denotes the number of edge-selected frames in $\mathcal{F}$. The normalization alleviates the scale mismatch between spatial grids and frame indices, making the subsequent attention-shift metric more stable. This 3-D position preserves spatial geometry within each frame while explicitly encoding temporal ordering across frames.

Given the visual attention distribution A_l^{vis} of the l th draft token, we compute its attention centroid as

$$X_l^c = \frac{\sum_j \hat{u}_j A_l(j)}{\sum_j A_l(j)}, Y_l^c = \frac{\sum_j \hat{v}_j A_l(j)}{\sum_j A_l(j)}, Z_l^c = \frac{\sum_j \hat{t}_j A_l(j)}{\sum_j A_l(j)} \quad (13)$$

where (X_l^c, Y_l^c) captures the spatial attention focus within a frame, while Z_l^c reflects the dominant frame selected by the model.

The spatio-temporal shift of attention is measured by the 3-D Euclidean distance between the centroid of the current iteration and its parent node in the speculative tree

$$\Delta A = \frac{1}{k} \sum_l \sqrt{(X_l^c - X_l^{\text{prev}})^2 + (Y_l^c - Y_l^{\text{prev}})^2 + (Z_l^c - Z_l^{\text{prev}})^2} \quad (14)$$

where X_l^{prev} , Y_l^{prev} , and Z_l^{prev} denote the centroid coordinates of the parent node of token l in the speculation tree.

Then, based on the distance, the shift decision is calculated as

$$R_a = \begin{cases} 1, & \text{if } \Delta A \geq \tau_a \\ 0, & \text{if } \Delta A < \tau_a \end{cases} \quad (15)$$

where τ_a denotes the attention shift threshold.

Finally, the routing module M_r integrates decisions from both aspects via the logical OR operation, yielding the final output

$$R = R_d \vee R_a. \quad (16)$$

If either confidence is insufficient ($R_d = 1$) or the attention shift is detected ($R_a = 1$), the original model is invoked for verification; otherwise, the generation proceeds uninterrupted.

While the edge has already filtered out irrelevant frames, the visual context $\mathcal{F}$ is highly focused on the semantics of the video. Thus, attention shifts are more likely to reflect genuine semantic transitions rather than noise, enhancing the routing accuracy.

B. Tree-Based Parallel SPD

To provide more candidate branches during verification, we introduce a tree-based parallel draft generation and verification mechanism. Multiple draft sequences are drafted and verified in parallel, as shown in Fig. 4. To support parallel draft and verification generation, we design specialized position encodings, hierarchy-aware attention masks, and a tree-structured KV-cache. Under identical input prefixes, these designs strictly preserve the autoregressive dependency structure of each candidate branch, ensuring that the resulting token prediction distributions are identical to those produced by standard autoregressive decoding, while significantly accelerating inference via multisequence parallel execution.

1) *Tree-Based Parallel Draft Generation*: It is given as follows.

a) *Draft flattening*: To facilitate merging and transmission, multiple draft sequences are organized in a tree structure and flattened into a 1-D sequence for parallel computation. Given the current draft tree T_d , it consists of verified tokens n_0^d and draft tokens $n_{1,1}^d, n_{1,2}^d, \dots, n_{l,k}^d$, where l denotes the depth of the deepest layer and k denotes its number of branches. The flattened sequence is constructed as $S_{flat}^d = [n_0^d, n_{1,1}^d, n_{1,2}^d, \dots, n_{l,k}^d]$.

b) *Parallel generation*: For efficient tree-oriented parallel computation, we introduce customized position IDs and attention masks to support hierarchy-aware draft generation. Tokens at the same tree depth share identical position IDs, while the attention masks ensure that each token only attends to earlier tokens in its own branch, preventing cross-branch interaction. We further adopt a tree-structured KV-cache to maximize reuse and eliminate redundant computation.

c) *Tree-oriented merging*: Using the flattened sequence S_{flat}^d , the lightweight draft model generates k logits in parallel from the k tokens at layer l , which are then merged according to their hierarchical relationships. Various merging strategies can be applied here, such as confidence-based selection, and we adopt a simple Top- q selection; each draft node spawns

q child branches using the Top- q logits. This hierarchical organization enables the model to explore multiple candidate paths efficiently while avoiding redundant computation.

d) *Tree pruning*: Under adaptive verification, the intervals between verification steps are dynamic, causing the draft tree to grow with variable depth. Although this does not immediately increase inference cost, an excessively deep or dense tree can eventually degrade latency [64]. To prevent this, given the draft tree T_d and a maximum depth threshold L , once the depth reaches $l \geq L$, we prune the tree by retaining only the highest confidence branch (i.e., the left subtree). While multiple pruning strategies (e.g., confidence-based or access-frequency-based) are possible, we use the simplest approach. All nodes preceding the pruned point are assumed correct and will no longer undergo verification or rollback.

2) *Tree-Based Parallel Draft Verification*: It is given as follows.

a) *Verify flattening*: To facilitate parallel verification, the hierarchical structure of the draft tree must be flattened into a single sequence. Unlike the draft generation phase, where only speculative tokens are considered, the verification process includes both the verified tokens from previous iterations (e.g., n_0^d) and the current speculative tokens. Given the current draft tree T_d , which consists of verified tokens n_0^d and speculative tokens $n_{1,1}^d, n_{1,2}^d, \dots, n_{l,k}^d$, the flattened sequence for verification is constructed as $S_{flat}^v = [n_0^d, n_{1,1}^d, n_{1,2}^d, \dots, n_{l,k}^d]$.

b) *Parallel verification*: During parallel verification, the original model evaluates the flattened sequence S_{flat}^v . Tokens at the same tree depth share identical position IDs to preserve their relative hierarchical roles, and a hierarchy-aware attention mask ensures that each token attends only to preceding tokens within its own branch. The verification process also leverages the KV-cache to reuse cached key-value pairs for previously processed tokens, reducing redundant computations.

c) *Tree-oriented selection*: Through parallel verification, we obtain two trees, which are the draft tree T_d and the verification tree T_v . Since T_v is constructed based on T_d , each node in T_d has a corresponding predecessor in T_v . However, because multiple branches coexist in T_d , the final accepted sequence cannot be determined from T_v alone, requiring both trees to work jointly to identify the correct path.

To recover the accepted sequence, we adopt a tree-oriented selection mechanism. As described in Algorithm 2, the selection process begins at the root of the draft and verify tree. For a given confirmed parent node u , the target model computes the probability distribution for the next token. We select the target token t_{target} via greedy argmax-selection.

- i) *Acceptance*: The algorithm check if t_{target} exists among the immediate children of u in the draft tree T_d . If a child node c satisfies $c = t_{target}$, the draft is accepted. The system then advances to node c and repeats the verification for its children.
- ii) *Rejection and Fallback*: If no child node matches t_{target} , the speculative path is invalid. The process terminates immediately, and the mismatched t_{target} is appended to the final sequence as the correction token. All subsequent tokens in the draft branch are discarded. This mechanism guarantees that the final output sequence S

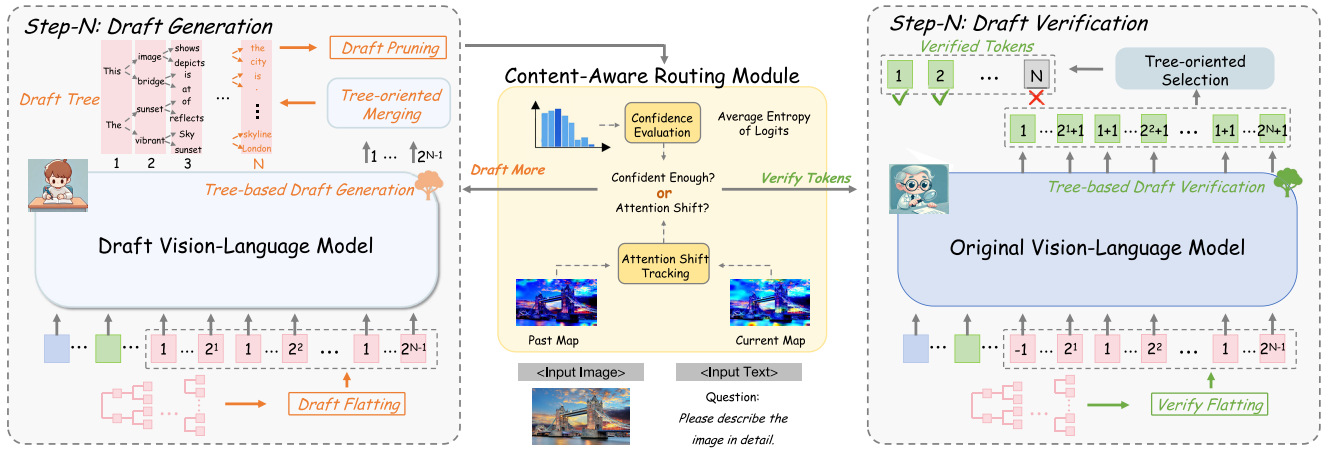


Fig. 3. Cloud MLLM speculative acceleration. In contrast to the vanilla fixed-window SPD, our SEC adaptively routes the draft generation and verification, based on the confidence scores and attention distributions. Furthermore, SEC leverages tree-based parallel decoding to provide more candidates for improving speculative acceptance (best viewed in color).

Algorithm 2 Tree-Oriented Verification and Selection

Input: Draft Tree T_d , Verify Logits Tree T_v
Output: Final Accepted Sequence S
Initialize $S \leftarrow []$;
Set current draft node $u \leftarrow \text{root}(T_d)$;
Set current verify logits $v \leftarrow \text{suffix logits of } u \text{ in } T_v$;
while True **do**
 $t_{\text{target}} \leftarrow \text{argmax}(v)$; // Greedy Selection
 Append t_{target} to S ;
 if u has child c in T_d and $c == t_{\text{target}}$ **then**
 % Accept: Move down the tree
 $u \leftarrow c$;
 $v \leftarrow \text{logits for } c \text{ in } T_v$;
 else
 % Reject and Fallback: Stop verification
 Break;
Return S ;

is mathematically identical to one generated by the target model in isolation, as any deviation by the draft model is instantly corrected by the target model's prediction.

3) *Topology-Aware Parallel Attention Computation:* To ensure that our tree-based decoding mathematically replicates the standard autoregressive process, we rigorously control the attention computation via a topology-aware mask and depth-based position indices.

In standard auto-regressive decoding, the attention is computed as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M_a\right)V \quad (17)$$

where M_a is a strict lower triangular causal mask, and d_k is the query-key dimension.

Conceptually, the draft tree T_d represents a bundle of multiple candidate sequences $\mathcal{S} = \{S_1, S_2, \dots, S_m\}$, where each sequence S_i corresponds to a distinct path from the root to a leaf node. Validating the tree is mathematically equivalent to validating all sequences in $\mathcal{S}$ simultaneously. However, treating them as independent batch items causes significant memory redundancy due to the shared prefixes (common

ancestors). To eliminate this redundancy while enabling parallel verification, we flatten the unique nodes of T_d into a single 1-D sequence S_{flat} , and reconstruct the logical independence of each sequence S_i via topology-aware attention mechanisms.

As shown in Fig. 4, to ensure that the verification of each candidate sequence S_i within the flattened buffer S_{flat} remains strictly isolated from irrelevant branches, we construct a tree attention mask M^{tree} . Unlike the standard causal mask, which allows attending to all prior tokens, M^{tree} enforces the tree topology constraints. For any two tokens $u, v \in S_{\text{flat}}$, the attention score is masked as

$$M_{u,v}^{\text{tree}} = \begin{cases} 0, & \text{if } v \in \text{prefix}(u) \cup \{u\} \\ -\infty, & \text{otherwise.} \end{cases} \quad (18)$$

This mechanism guarantees that, when computing the representation of a token on branch A , the model is strictly prevented from attending to any tokens on branch B . Consequently, the model perceives distinct logical sequences in parallel within a single forward, ensuring that the output logits for each node are identical to those produced by standard auto-regressive decoding.

What is more, standard sequential position IDs are insufficient for the nonlinear structure of T_d . To preserve the correct relative distances for positional embeddings (e.g., RoPE) across multiple branches, we assign position IDs based on the hierarchical depth rather than the linear index in S_{flat} . Specifically, as shown in Fig. 4, for a token u at depth l in the tree, its position ID is assigned as

$$\text{PosID}(u) = \text{PosID}(\text{root}(u)) + l \quad (19)$$

where $\text{root}(\cdot)$ denotes the root node of the corresponding draft sequence, whose position ID equals the length of the input prompt. This ensures that tokens at the same depth across different candidate sequences share the same positional context, maintaining geometric consistency with the standard auto-regressive decoding.

4) *Tree Implementation and Complexity Analysis:* The computational overhead of our cloud-side tree-based parallel decoding can be analyzed purely from the perspective of tree

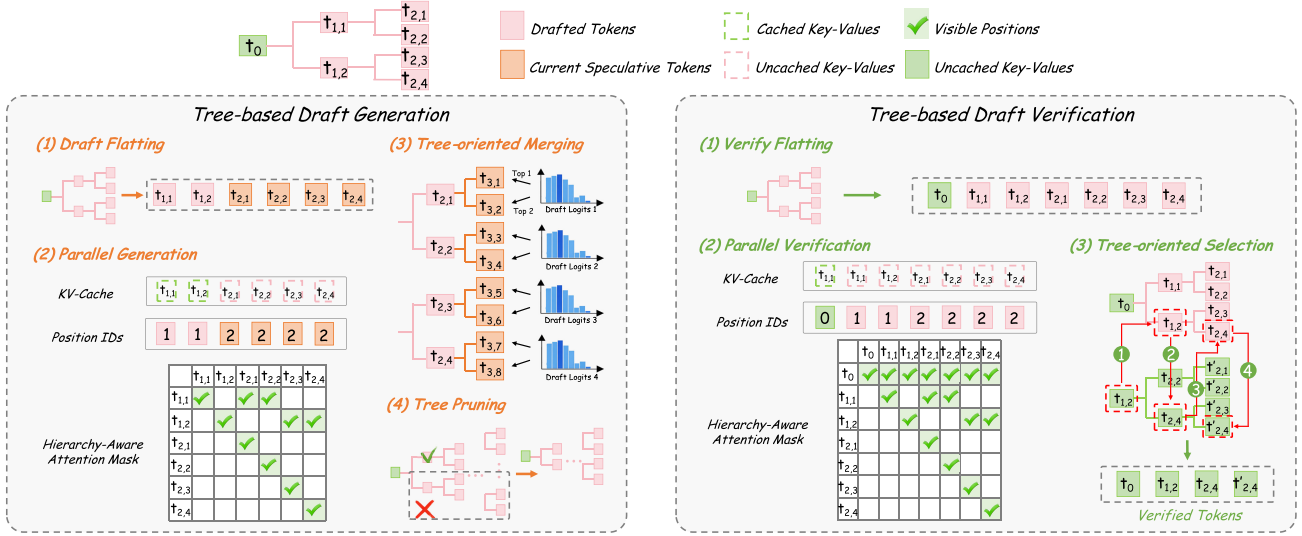


Fig. 4. Tree-based parallel decoding on the cloud. To provide more candidate branches during verification, *SEC* employs a tree-based decoding mechanism on the cloud to enhance parallelism.

operations. Given the draft tree with maximum depth L and branches per node q , the theoretical upper bound on the total number of possible nodes is $O(q^L)$. Importantly, this quantity characterizes only the maximum tree size, rather than the cost of any tree operation. To avoid explicitly constructing or traversing an exponentially sized structure, we store the tree in a preallocated 1-D buffer with a fixed tree-to-sequence mapping in breadth-first order, as shown in (20). Note that we separate the overhead of tree-structure maintenance from model-side computation, and only analyze the former here

$$\left[\begin{array}{c} \underbrace{n_0}_{\text{root}}, \underbrace{n_{1,0}, \dots, n_{1,q-1}}_{\text{depth 1}}, \underbrace{n_{2,0}, \dots, n_{2,q^2-1}}_{\text{depth 2}}, \\ \dots, \underbrace{n_{L,0}, \dots, n_{L,q^L-1}}_{\text{depth } L} \end{array} \right]. \quad (20)$$

With this fixed layout, both drafting and verifying can directly consume the maintained 1-D sequence with customized position IDs and attention masks, thus introducing no explicit flattening overhead. Similarly, with the newly generated tokens, tree-oriented merging can be directly achieved by attaching the tokens to the sequence, eliminating any explicit merging overhead. Since the mapping from tree to the buffer sequence is fixed, pruning can be performed using a fixed slicing template that retains only the highest confidence branch, at most L times, with the complexity $O(L)$. To recover the final accepted sequence, our tree-oriented selection jointly traverses the verification tree T_v and the draft tree T_d , matching the verified token against at most q candidates at each level, which results in $O(L \cdot q)$.

Therefore, under the 1-D buffer representation, the structural maintenance overhead of flattening, merging, pruning, and tree-oriented selection scales linearly with the depth L and branching factor q of the tree, i.e., $O(L \cdot q)$, and no tree operation incurs exponential complexity. Furthermore,

depth-bounded pruning keeps the draft tree within a strictly controlled range, ensuring that even when the tree reaches its maximum size, structural overhead remains lightweight and does not introduce uncontrolled latency into decoding.

C. Theoretical Speedup Analysis

To quantify the expected efficiency gains of our adaptive SPD, we provide a simplified analytical model. We consider the common deployment regime where the model inference operates below GPU saturation, such that the wall-clock verification latency is dominated by kernel launch and memory access overhead rather than token count. Let C_v be the time cost for the large cloud model to verify a sequence, and C_d be the cost for the small model to draft one token. C_r indicates the relatively small routing overhead. Typically, $C_r \ll C_d \ll C_v$.

When generating n tokens, the standard autoregressive decoding requires time $C_{\text{auto}} = n \cdot C_v$. In our framework, the system incurs a drafting cost of $D \cdot C_d$, a relatively small routing overhead C_r , and a single verification cost C_v , with average draft depth D . Let $\mathbb{E}_n$ correspond to the expected length of the longest prefix of the target sequence that can be matched within the draft tree during a single verification step. The total time to generate L tokens is approximately

$$C_{\text{ours}} \approx \frac{n}{\mathbb{E}_n} (D \cdot (C_d + C_r) + C_v). \quad (21)$$

The theoretical speedup ratio is thus

$$\begin{aligned} \text{Speedup} &= \frac{C_{\text{auto}}}{C_{\text{ours}}} = \frac{\mathbb{E}_n \cdot C_v}{D \cdot (C_d + C_r) + C_v} \\ &= \frac{\mathbb{E}_n}{1 + D \cdot ((C_d + C_r)/C_v)}. \end{aligned} \quad (22)$$

Our tree-based decoding (Section V-B) boosts $\mathbb{E}_n$ by verifying parallel branches, while the lightweight draft model and adaptive routing keep the overhead term $D \cdot ((C_d + C_r)/C_v)$ negligible. Furthermore, adaptive routing (Section V-A) reduces the frequency of expensive verifications (C_v), enhancing speedup beyond the static bound.

VI. RESULTS AND DISCUSSION

In this section, we present a comprehensive evaluation of our proposed semantic-aware edge–cloud collaborative framework for low-latency video understanding.

A. Experiment Setup

1) *Datasets*: In our experiments, we leverage LMMsEval to evaluate the performance of our method. Three widely used video benchmarks, VideoMME [39], LongVideoBench [40], and Perception Test [41], are used for evaluation. All datasets and corresponding splits follow the configurations that are publicly available on the LMMsEval repository in Hugging Face.

VideoMME [39] is a comprehensive benchmark designed to assess multimodal reasoning capabilities over video content. It features diverse tasks that require both visual understanding and language reasoning, making it suitable for evaluating models' ability to capture complex semantic relationships in dynamic scenes. Note that we evaluate VideoMME without subtitles.

LongVideoBench [40] focuses on evaluating models' ability to understand long-form videos, typically several minutes in duration. The evaluation tasks include question answering, event prediction, and summary generation, all based on long video sequences.

Perception Test [41] is a novel multimodal video benchmark designed to evaluate the perception and reasoning skills of pretrained MLLMs. We use the MC-vQA task of the benchmark in our evaluation, including short-horizon tasks such as change detection, action & sound recognition, and object, action & event counting, etc. It is well-suited for evaluating models' performance on short-horizon streaming tasks that require precise spatial and temporal localization/understanding of video content.

In addition to these public datasets, we further construct a customized benchmark named VideoMME-CP, built upon the original VideoMME dataset, specifically targeting short-horizon perception abilities. It focuses on six representative task types from VideoMME: Counting Problem, Spatial Perception, Spatial Reasoning, Temporal Perception, and Temporal Reasoning. The benchmark contains a total of 610 questions across 213 videos, providing a fine-grained and more diagnostically focused subset for controlled evaluation.

2) *Implementation Details*: In our experiments, we evaluate the effectiveness of SEC using several state-of-the-art VLM families as baselines: Qwen2-VL [18], InternVL2 [19], and LLaVA-OneVision [77]. We designate the large-sized models (e.g., 7B-26B) as the original models and the small-sized models (e.g., 1-2B) as the draft models. All the pretrained weights of the VLMs were directly obtained from their HuggingFace repositories. During preprocessing, videos are decoded at the frame rate of 1 FPS with ffmpeg and divided into $M = 64$ segments, after which our semantic-aware frame selection module identifies N key frames per video based on query relevance and diversity.

Since our SEC is fully deterministic and does not rely on any stochastic operations, the accuracy, latency, and

transmission metrics exhibit no run-to-run variance. To ensure reproducibility, all experiments are conducted with fixed seeds of 2025 across PyTorch, NumPy, and CUDA. The frame-query matching scores are computed using TinyCLIP [81], a lightweight VLM well-suited for edge deployment. As shown in Table II, for frame selection on the edge, we set the tradeoff factors α and γ as 0.8, prioritizing query alignment while preserving sufficient temporal spread. The initial segment number M is set to 64. For cloud-side SPD, the initial shift threshold $\tau_d^{initial}$ and the upper limit $\hat{\tau}_d$ are set to 0.4 and 5, respectively. The adjustment learning rate lr is set to 0.1. These values provide a conservative initialization that stabilizes early decoding stages, and a reasonable ceiling to prevent the threshold from drifting to overly permissive regimes under low-entropy segments. We configure the maximum depth L and branch number q of the draft tree to 6 and 3, respectively. The attention shift threshold is set to $\tau_a = 0.8$. More insights and extended analyses have been relocated to Section VI-G.

During validation, we follow the default system prompts provided by the LMMs-Eval benchmark for each model family to ensure fair and standardized evaluation. On top of these defaults, we add a lightweight and consistent answer-format constraint across all models to reduce formatting variance: "The answer to this question is: {answer}." This minimal modification ensures consistent output formatting without influencing the reasoning process of the model.

The entire framework is implemented in Python using PyTorch 2.3.1 and Hugging Face Transformers 4.45.0. To simulate a realistic edge–cloud collaboration scenario, we deploy the framework components on heterogeneous hardware platforms that reflect practical deployment constraints. The edge side is simulated on a server equipped with an NVIDIA GeForce RTX 3060 12-GB GPU, an Intel Core i7-12700 CPU, and 16-GB RAM. This configuration represents a mid-tier edge server commonly found in IoT and smart camera systems, where computational resources are limited, and energy efficiency is critical. The cloud server, in contrast, is configured with 4×NVIDIA L20 48GB GPUs, 64 CPU cores, and 1024GB of RAM, providing abundant computational power for running large-scale MLLMs on the cloud.

To precisely account for communication overhead, we further model the edge–cloud link using realistic transmission characteristics. We evaluate three representative network configurations to capture a wide range of deployment environments. The first configuration corresponds to a high-bandwidth but higher latency backbone (50-ms latency and 500-Mb/s bandwidth). The second configuration reflects a more balanced setting (20-ms latency and 100-Mb/s bandwidth). The third configuration models a low-latency but bandwidth-constrained link (5-ms latency and 50-Mb/s bandwidth). This setup enables a comprehensive evaluation of end-to-end latency, bandwidth consumption, and system scalability under diverse and practically relevant edge–cloud communication conditions.

B. End-to-End Response Latency Evaluation

To understand the practical benefits of our proposed SEC framework in realistic transmission and inference conditions, we conduct an end-to-end latency analysis under three typical

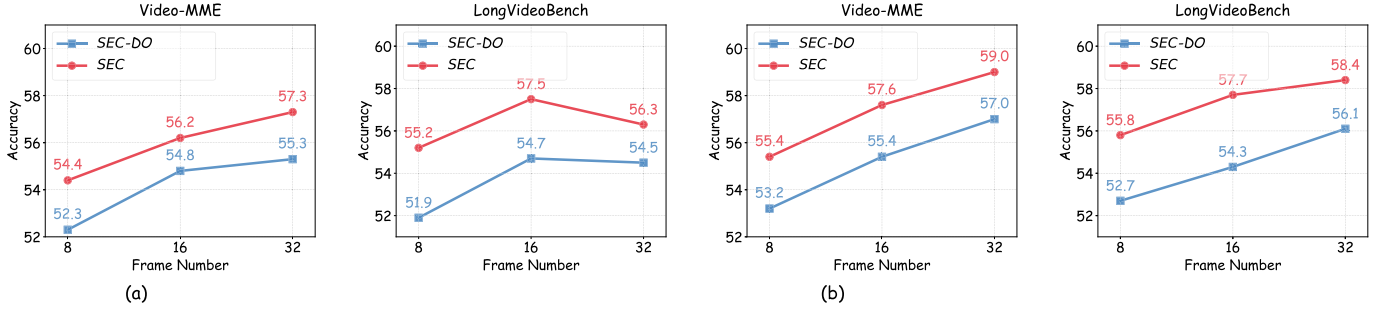


Fig. 5. Performance comparison between *SEC*-Decoding only (*SEC*-DO, i.e., cloud-side adaptive SPD with uniform frame sampling), and our *SEC* across different model scales and benchmarks. We evaluate the VideoMME and LongVideoBench scores under varying numbers of input frames (8, 16, and 32), with InternVL2-2B as the speculative draft model. Our method consistently outperforms uniform sampling across all settings, achieving higher accuracy with fewer frames.

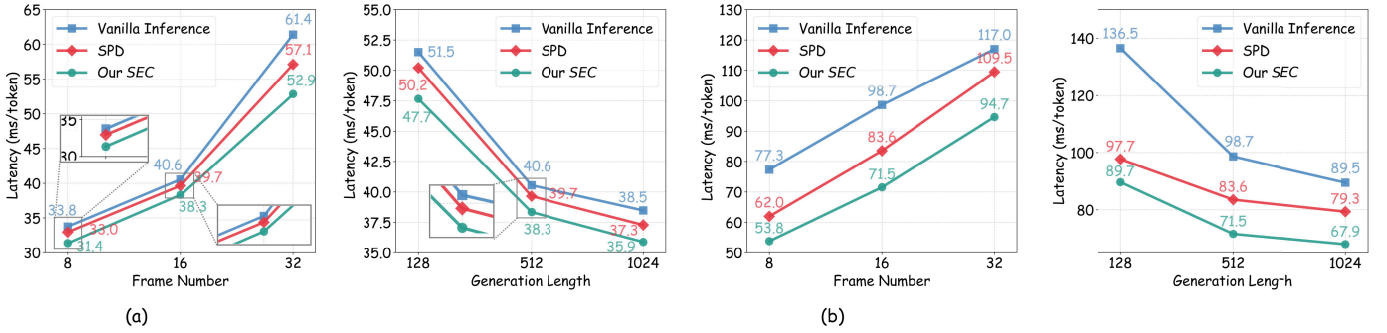


Fig. 6. Latency comparison on transmission link 2 (20-ms latency and 100-Mb/s bandwidth). We compare three inference paradigms: vanilla inference, *SEC*-selection only (*SEC*-SO, i.e., cloud-side standard SPD with our edge-side semantic frame selection), and our full *SEC* framework. Per-token latency (ms/token) is evaluated under varying input frame counts (8, 16, and 32 frames) and output lengths (128, 512, and 1024 tokens), using InternVL2-2B as the draft model and InternVL2-26B as the target model. Our *SEC* achieves significantly lower latency than both Vanilla and SPD across all configurations, demonstrating its efficiency advantage without sacrificing accuracy.

TABLE II
DETAILED HYPERPARAMETER CONFIGURATION

Edge-side Selection			Cloud-side Decoding						
M	α	γ	$\tau_d^{initial}$	$\hat{\tau}_d$	τ_a	lr	L	q	
64	0.8	0.8	0.4	5	0.8	0.1	6	3	

network link settings, which are a high-bandwidth link 1 (50-ms latency and 500-Mb/s bandwidth), a balanced link 2 (20-ms latency and 100-Mb/s bandwidth), and a low-latency link 3 with reduced bandwidth (5-ms latency and 50-Mb/s bandwidth).

As shown in Table III, we compare the proposed *SEC* (8 input frames) with vanilla inference (16 input frames) under matched accuracy conditions (as shown in Fig. 5). Vanilla inference denotes uniform frame sampling performed on the edge side, followed by standard autoregressive decoding on the cloud side. The per-token latency is measured over the generation of 256 output tokens. Across all link types and model scales, *SEC* consistently reduces end-to-end per-token latency of 2.1 ms/token to 4.4 ms/token in the 8B setting and 33.8 ms/token to 35 ms/token in the 26B setting, using only half the input frames. Crucially, this efficiency gain does not come at the cost of performance. *SEC* achieves comparable or even higher scores on both VideoMME and LongVideoBench,

demonstrating that the selected eight semantically salient frames preserve discriminative visual-temporal information more effectively than uniformly sampled frames. These results confirm that *SEC* enables efficient multimodal inference under diverse link status.

C. Accuracy Evaluation With State of the Arts

We evaluate the end-to-end accuracy of *SEC* on three prominent video benchmarks: VideoMME [39], LongVideoBench [40], and Perception Test [41].

1) *General Video Tasks*: As summarized in Table IV, we compare against a broad spectrum of state-of-the-art models with our *SEC*, including both general-purpose MLLMs and specialized frame selection methods (Selector [35], Q-Frame [36], and AKS [37]). Across all models, including LLaVA-OV-7B, Qwen2-VL-7B, InternVL2-8B, and InternVL2-26B, *SEC* consistently boosts the accuracy by 1.6%–2.0%. Among them, InternVL2-26B achieves 59.0% on VideoMME with *SEC*, surpassing its baseline of 57.0% and other state-of-the-art video MLLMs such as MiniCPM-V2.6 with 60.9% and ShareGPT4Video with 39.9%. Compared with recently competitive sampling methods, Selector [35], Q-Frame [36], and AKS [37], our *SEC* likewise improves video understanding through more effective and adaptive frame selection. Importantly, *SEC* achieves these gains without increasing the

TABLE III

END-TO-END PER-TOKEN LATENCY COMPARISON ACROSS TRANSMISSION LINK CONFIGURATIONS. VANILLA INFERENCE DENOTES UNIFORM FRAME SAMPLING PERFORMED ON THE EDGE SIDE, FOLLOWED BY STANDARD AUTOREGRESSIVE DECODING ON THE CLOUD SIDE. WE COMPARE THE PROPOSED SEC (8 INPUT FRAMES) WITH VANILLA INFERENCE (16 INPUT FRAMES) UNDER MATCHED ACCURACY CONDITIONS (SEE FIG. 5). WE REPORT THE MEAN AS WELL AS THE 50TH AND 90TH PERCENTILES OF PER-TOKEN LATENCY ACROSS ALL VIDEOS AND QUERIES, OVER THE GENERATION OF 256 OUTPUT TOKENS. THREE TRANSMISSION LINK CONFIGURATIONS ARE EVALUATED: HIGH-BANDWIDTH LINK 1 (50 MS AND 500 MB/S), BALANCED LINK 2 (20 MS AND 100 MB/S), AND LOW-LATENCY LINK 3 (5 MS AND 50 MB/S). THE CONTRIBUTION OF EDGE SELECTION AND CLOUD DECODING IS FURTHER DISENTANGLED IN FIGS. 5 AND 6

Methods	Type	Response Latency (mean/P50/P90, ms/token)			Video Benchmark	
		High-bandwidth Link 1	Balanced Link 2	Low-latency Link 3	VideoMME (%)	LongVideoBench (%)
$\mathcal{M}_o$ InternVL2-8B [19]	Vanilla Inference	43.0 / 43.7 / 44.2	44.5 / 45.2 / 45.7	46.7 / 47.4 / 47.8	54.8	54.7
$\mathcal{M}_d$ InternVL2-2B [19]	+ SEC	40.9 / 41.1 / 41.5	41.4 / 41.5 / 41.9	42.3 / 42.4 / 42.6	54.4	55.2
$\mathcal{M}_o$ InternVL2-26B [19]	Vanilla Inference	107.4 / 108.2 / 108.7	108.9 / 109.6 / 110.1	111.1 / 111.8 / 112.4	55.4	54.3
$\mathcal{M}_d$ InternVL2-2B [19]	+ SEC	73.6 / 74.1 / 74.4	73.9 / 74.4 / 74.7	77.3 / 77.8 / 78.0	55.4	55.8

TABLE IV

ACCURACY COMPARISON WITH STATE-OF-THE-ART METHODS ON VIDEO-MME AND LONGVIDEOBENCH BENCHMARKS. WE EVALUATE VARIOUS VIDEO LLMs IN TERMS OF ACCURACY (%). VALUES IN BRACKETS DENOTE 95% CONFIDENCE INTERVALS ESTIMATED FROM 1000 BOOTSTRAP ITERATIONS. RESULTS SHOW THAT SEC BRINGS SIGNIFICANT IMPROVEMENTS ACROSS MODEL SCALES AND BENCHMARKS

Methods	Frames N	General Benchmark (Acc., 95% CI)	
		VideoMME (%)	LongVideoBench (%)
Video-LLaVA [23]	8	39.9	39.1
MiniCPM-V2.6 [82]	64	60.9	54.9
VideoLLaMA2 [76]	16	47.9	-
PLLaVA [78]	32	-	53.2
ShareGPT4Video [73]	-	39.9	41.8
LLaVA-OV-7B [77]	32	58.7 [56.4, 59.8]	56.4 [52.9, 58.1]
+ SEC	32	60.3 [58.4, 61.8]	58.9 [55.7, 60.2]
InternVL2-8B [19]	32	55.3 [52.9, 57.6]	54.5 [53.0, 57.9]
+ SEC	32	57.3 [55.8, 59.1]	56.3 [54.1, 58.4]
InternVL2-26B [19]	32	57.0 [55.3, 59.4]	56.1 [53.8, 58.2]
+ SEC	32	59.0 [57.5, 61.1]	58.4 [55.9, 59.7]
Qwen2-VL-7B [18]	32	57.3 [56.1, 59.0]	57.4 [55.8, 59.1]
+ Selector [35]	8	58.7 [-, -]	- [-, -]
+ Q-Frame [36]	44	58.3 [56.8, 59.5]	58.4 [56.7, 59.4]
+ AKS [37]	32	58.9 [57.3, 59.8]	59.4 [57.2, 61.1]
+ SEC	32	59.2 [57.8, 61.0]	59.5 [57.4, 61.4]

TABLE V

EVALUATION OF OUR SEC ON SHORT-HORIZON STREAMING TASKS. WE EVALUATE OUR SEC WITH VARIOUS MLLMs ON VIDEO-MME-CP AND PERCEPTIONTEST-MC, SHORT-HORIZON STREAMING TASK DATASETS. OUR PROPOSED SEC ENHANCES EACH BASE MODEL, SHOWING ITS EFFECTIVENESS AND GENERALIZABILITY

Methods	Frames N	Short-horizon Benchmark	
		VideoMME-CP (%)	PerceptionTest-MC (%)
Qwen2-VL-7B [18]	16	43.4	60.1
+ SEC	16	45.6	62.7
InternVL2-8B [19]	16	43.7	58.2
+ SEC	16	46.0	60.9
InternVL2-26B [19]	16	44.6	60.4
+ SEC	16	47.1	63.3

number of input frames or model size, highlighting its ability to extract more informative content from fewer inputs.

2) *Short-Horizon Streaming Tasks*: To further evaluate SEC's applicability in real-time streaming scenarios, we test

on two short-horizon benchmarks, simulating low-latency, short-horizon video understanding tasks. Table V, SEC consistently improves all base models by 2.2%–2.5% on VideoMME-CP and 2.6%–2.9% on PerceptionTest-MC, across varying model scales, e.g., 8B and 26B. In summary, the consistent gains across model sizes from 7B to 26B and task granularities of long-form and short-horizon strongly support SEC's generalizability, scalability, and deployment readiness. It serves as a plug-and-play enhancement that elevates both accuracy and efficiency, bridging the gap between theoretical model capacity and practical video-language understanding under resource constraints.

D. Performance Evaluation Across Input Frame Counts and Generation Lengths

To comprehensively evaluate the efficiency and effectiveness of our SEC, we conduct comparisons across varying input frame numbers (8, 16, and 32 frames) and generation lengths (128, 512, and 1024 tokens).

1) *Accuracy Comparison of SEC-Decoding Only and SEC Across Frame Counts*: As shown in Fig. 5, SEC consistently outperforms SEC-Decoding only (cloud-side adaptive SPD with uniform frame sampling) across both VideoMME and LongVideoBench benchmarks. Notably, a cross-frame comparison reveals that our approach enables sparse-frame inputs to surpass dense-frame baselines. Sparse-frames input with only 8 frames under our method achieves comparable or even superior performance to the baseline with 16 frames, while 16 frames under our method can outperform the baseline with 32 frames. This demonstrates that our semantic-aware sampling effectively captures task-relevant visual content, significantly improving information density per frame. By focusing on temporally and semantically salient moments, we reduce redundancy and enable more efficient use of bandwidth and computational resources, which are critical advantages in edge-cloud video analysis systems.

2) *Latency Comparison With Vanilla Inference and SEC-Selection Only on Transmission Link 2*: Fig. 6 presents end-to-end per-token latency comparisons on transmission link 2 (20-ms latency and 100-Mb/s bandwidth). We compare three inference paradigms: vanilla inference, SEC-selection

TABLE VI

IMPACT OF GENERATION LENGTH ON OUTPUT QUALITY. WE EVALUATE OUR *SEC* FRAMEWORK ON VIDEO-MME WITH LONG SEQUENCE CAPTIONING QUESTIONS, REPORTING ACROSS LENGTHS OF 256, 512, AND 1024 TOKENS. CONSISTENT SCORES ACROSS LENGTHS OF EACH MODEL DEMONSTRATE THE EFFECTIVENESS AND GENERALIZABILITY OF *SEC*

Methods	Generation Length	Generation Quality Assessment Metrics				
		Norm. Levenshtein	SBert Simi.	BertScore F1	Answer Consistency	FactScore
$\mathcal{M}_o$ InternVL2-8B [19]	256	0.1235	0.9184	0.9313	0.7743	0.9026
	512	0.1312	0.9163	0.9262	0.7699	0.9133
	1024	0.1297	0.9156	0.9243	0.7711	0.9105
$\mathcal{M}_d$ InternVL2-2B	256	0.1972	0.8798	0.9032	0.7312	0.8801
	512	0.1994	0.8853	0.9155	0.7278	0.8749
	1024	0.2002	0.8812	0.9003	0.7256	0.8874

only (*SEC*-SO, i.e., cloud-side standard SPD with our edge-side semantic frame selection), and our full *SEC* framework. Across all configurations, *SEC* achieves significantly lower latency than Vanilla inference and *SEC*-selection only on both InternVL2-8B and InternVL2-26B. This makes our *SEC* uniquely suited for real-time streaming applications where low latency and high throughput are non-negotiable.

3) *Generation Length Stability*: A key challenge in generation is maintaining factual consistency and semantic coherence as output length increases, especially for long-form captioning or explanatory QA. To rigorously assess whether *SEC* introduces or mitigates hallucination in long-form generation, a common concern with aggressive frame compression, we conduct a controlled comparison between vanilla inference and *SEC*-enhanced inference under identical generation lengths (256–1024 tokens) on long-sequence captioning questions from VideoMME. Specifically, we use five complementary automatic metrics: normalized Levenshtein distance, Sentence-BERT Similarity, BERTScore F1, Answer Consistency, and FactScore.

As shown in Table VI, *SEC* consistently yields higher or comparable scores across all metrics, even at extended generation lengths of 1024 tokens. This demonstrates strong alignment with the outputs of the vanilla inference, indicating that *SEC* does not introduce additional semantic drift or hallucinatory degradation as output length increases. Instead, its query-driven frame selection preserves the core visual evidence required for faithful reasoning, ensuring that the generated responses remain stable.

E. Edge-Side Processing and Transmission Cost

To assess the practicality of *SEC* in real-world streaming applications, we evaluate its edge and link cost under varying spatiotemporal constraints, the number of input frames (8, 16, and 32), and output generation lengths (256, 512, and 1024 tokens), across model scales (InternVL2-8B and -26B, with InternVL2-2B as draft).

1) *Multistream Efficiency Comparison*: As shown in Table VII, we compare uniform sampling and our *SEC* under varying multistream concurrence (1, 2, and 4 streams) of 60 FPS on VideoMME, with all configurations calibrated to achieve equivalent accuracy on VideoMME and LongVideoBench (see Fig. 5). Specifically, the proposed *SEC* operates with 8 input frames, whereas vanilla inference

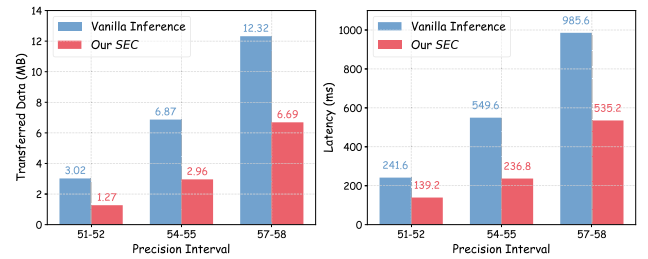


Fig. 7. Data efficiency and latency reduction of *SEC* versus Uniform Sampling under matched precision intervals. We compare transmission data (left) and inference latency (right) across three precision ranges (51%–52%, 54%–55%, and 57%–58%) on VideoMME. Our *SEC* consistently reduces both data transfer and latency while maintaining equivalent precision.

uses 16 frames. The results demonstrate that *SEC* substantially reduces edge-side processing demands and transmission overhead, without any accuracy tradeoff, attaining the same accuracy with only half the input frames. This ensures that the latency and bandwidth reductions are not achieved at the cost of degraded understanding, but rather stem purely from *SEC*'s more efficient use of visual information. Even under increasing concurrence, *SEC* maintains this parity while consistently cutting transmission data and end-to-end latency. These results establish *SEC* as a precision-preserving compression framework, enabling substantial edge-to-cloud efficiency gains while rigorously maintaining model fidelity.

2) *Edge-Side Computation Overhead Comparison*: To quantify the edge-side efficiency of our semantic frame selection module, Table VIII reports its deployment overhead under a unified protocol (batch size of 64 and input resolution of 224×224). We measure system-level metrics, including CPU/RAM usage, GPU utilization/memory, FLOPs, latency, and power.

As shown in Table VIII, large CLIP backbones (e.g., ViT-L/14 and ViT-B/32) provide strong representations but incur substantially higher compute and GPU overhead. In contrast, TinyCLIP variants consistently reduce resource consumption across all metrics. Despite being lightweight, the TinyCLIP encoder used in *SEC* preserves comparable performance on VideoMME and LongVideoBench, making it better suited for low-power and latency-sensitive edge deployment.

3) *Transmission Efficiency*: As illustrated in Fig. 7, we quantitatively evaluate the transmission data volume

TABLE VII

EDGE-SIDE COMPUTATION AND TRANSMISSION COST COMPARISON UNDER MULTISTREAM SETTINGS WITHIN THE MATCHED ACCURACY INTERVAL ON TRANSMISSION LINK 2. WE EVALUATE ONE, TWO, AND FOUR CONCURRENT STREAMS OF 60 FPS WITH $\mathcal{M}_o$ INTERNVL2-26B AND $\mathcal{M}_d$ INTERNVL2-2B ON VIDEO-MME. ALL CONFIGURATIONS ARE CALIBRATED TO ACHIEVE COMPARABLE ACCURACY ON VIDEO-MME AND LONGVIDEOBENCH (AS SHOWN IN FIG. 5). *SEC* INTRODUCES NEGLIGIBLE OVERHEAD FOR FRAME-QUERY MATCHING AND MARGINAL RELEVANCE COMPUTING, YET REDUCES TRANSMISSION DATA AND LATENCY ACROSS ALL SETTINGS, EFFECTIVELY DOUBLING THROUGHPUT

Streams 60 FPS	Methods	Frame	Frame-Query Matching Matching	Marginal Relevance Computing	Transmission Data	Transmission Latency Latency	Total Latency	Throughput
1	Uniform Sampling	16	0.0ms	0.0ms	6.43MB	514.4ms	514.4ms	1.94
	Our <i>SEC</i>	8	6.7ms	0.7ms	2.96MB	236.8ms	244.2ms	4.09
2	Uniform Sampling	16	0.0ms	0.0ms	12.86MB	1028.8ms	1028.8ms	0.97
	Our <i>SEC</i>	8	8.2ms	0.8ms	5.92MB	473.6ms	482.6ms	2.07
4	Uniform Sampling	16	0.0ms	0.0ms	25.72MB	2057.6ms	2057.6ms	0.49
	Our <i>SEC</i>	8	12.6ms	1.3ms	11.84MB	947.2ms	961.1ms	1.04

TABLE VIII

EDGE-SIDE COMPUTATIONAL COST OF OUR SEMANTIC FRAME SELECTION. WE COMPARE THE MODEL SIZE, MEMORY FOOTPRINT, FLOPS, POWER CONSUMPTION, AND OTHER DEPLOYMENT METRICS AT A BATCH SIZE OF 64, USING A STANDARD INPUT RESOLUTION OF 224×224

Methods	Param.	CPU Usage	RAM	GPU Usage	GPU Mem.	FLOPs	Latency	Avg. Power	Video Benchmark	
									VideoMME (%)	LVBench (%)
CLIP ViT-L/14	427.6M	15.80%	1377.4MB	95.30%	915.8MB	2418.2G	202.6ms	168.35w	57.8	58.0
CLIP ViT-B/32	151.3M	14.90%	896.3MB	64.60%	332.3MB	535.4G	24.1ms	126.51w	57.8	57.9
TinyClip ViT-40M/32	84.2M	10.80%	819.1MB	29.30%	194.7MB	236.1G	12.0ms	58.92w	57.6	57.8
TinyClip ViT-8M/16	23.5M	9.10%	767.5MB	26.80%	83.7MB	208.8G	8.3ms	41.68w	57.6	57.7

and latency across three matched precision intervals on VideoMME with $\mathcal{M}_o$ InternVL2-26B and $\mathcal{M}_d$ InternVL2-2B.

Our *SEC* consistently outperforms uniform sampling. For every precision tier, it reduces transmitted data by 50%–60% and inference latency by 40%–55%. Crucially, these gains are achieved without sacrificing accuracy. Each bar pair represents configurations calibrated to operate within the same precision bin, confirming that *SEC* delivers higher efficiency at identical performance levels.

As a result, the efficiency gains of *SEC*, which stem from the reduction in transmitted data volume, are agnostic to network volatility. Under more constrained or unstable network environments (e.g., lower bandwidth or higher latency), the reduced transmission demand directly translates into equal or larger relative latency savings, suggesting that the benefits of *SEC* would further amplify rather than degrade in more extreme network settings.

4) *Selected Frame Volume Comparison*: As shown in Fig. 8, we report the average amount of data transmitted (MB) when sending 2, 4, 8, 16, and 32 input frames, with error bars showing the standard deviation across trials. Across all settings, our *SEC* maintains nearly identical transmission costs to uniform sampling, demonstrating that semantic-aware frame selection does not introduce additional bandwidth overhead. Even though our *SEC* dynamically prioritizes informative frames, its data footprint remains comparable to uniform sampling, confirming its suitability for bandwidth-constrained video streaming.

5) *Accuracy–Latency Tradeoff Evaluation*: As shown in Table IX and Fig. 9, we compare random, uniform, similarity Top-N, and our *SEC* under the same bandwidth budget. Our *SEC* consistently achieves a superior accuracy–latency tradeoff across different frame budgets N , delivering higher

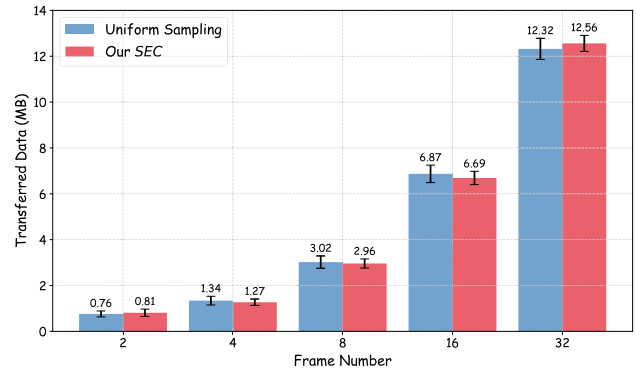


Fig. 8. Data volume comparison between uniform sampling and *SEC* under varying frame counts. We measure the average data transferred (MB) for 2, 4, 8, 16, and 32 input frames, with error bars indicating the standard deviation. Our *SEC* remains comparable data volume to uniform sampling across all frame counts, without any transmission overhead.

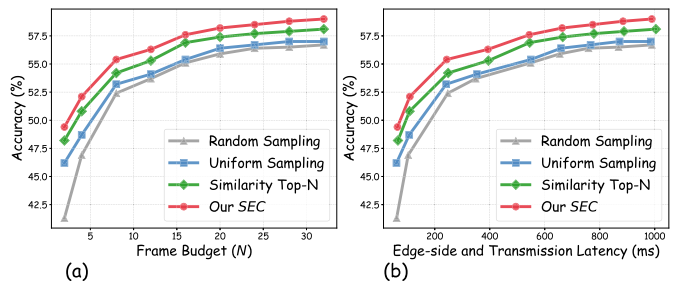


Fig. 9. Accuracy–latency tradeoff with matched bandwidth budgets. *SEC* achieves higher accuracy with lower latency compared to other alternatives.

accuracy at comparable edge-side computation and transmission costs. Table IX reports the fixed-budget setting, where

TABLE IX

COMPARISON OF EDGE AND TRANSMISSION OVERHEAD UNDER THE SAME BANDWIDTH BUDGET. WE EVALUATE RANDOM, UNIFORM, AND SIMILARITY TOP-N SAMPLING (SIM-TOPN) ALTERNATIVES AGAINST OUR *SEC* ON VIDEO-MME AND LONGVIDEOBENCH

Methods	Frames N	Video Benchmark		Data Volume	Latency
		VideoMME (%)	LVBench (%)		
Random	16	55.1	53.9	6.71MB	536.8ms
Uniform	16	55.4	54.3	6.87MB	549.6ms
Sim-TopN	16	56.9	56.1	6.73MB	545.3ms
Our <i>SEC</i>	16	57.6	57.7	6.69MB	544.1ms

TABLE X

INFERENCE LATENCY COMPARISON ON IMAGE ANALYSIS ACROSS DIFFERENT GENERATION LENGTHS ON TRANSMISSION LINK 2. THE GL DENOTES GENERATION LENGTH

Methods	Type	Latency(ms/token)		
		GL=128	GL=256	GL=512
$\mathcal{M}_o$ InternVL2-8B [19]	Vanilla	36.2	33.4	31.8
$\mathcal{M}_d$ InternVL2-2B [19]	SPD	34.7	32.3	31.4
	<i>SEC</i>	34.1	32.1	30.2
$\mathcal{M}_o$ InternVL2-26B [19]	Vanilla	88.4	79.5	75.2
$\mathcal{M}_d$ InternVL2-2B [19]	SPD	75.4	70.2	64.9
	<i>SEC</i>	51.9	51.1	49.5

SEC remains superior to all alternatives under identical resource constraints. This advantage stems from jointly considering semantic relevance and redundancy during candidate selection, which avoids wasting bandwidth on visually plausible but question-irrelevant frames. Overall, *SEC* provides a more robust and deployment-friendly frame selection policy, improving accuracy without increasing edge overhead.

F. More Results on Image Analysis

While our framework is primarily designed for video understanding in edge-cloud scenarios, we further evaluate its effectiveness on static image analysis tasks to demonstrate its generalizability across modalities.

We first examine the impact of *SEC* on inference latency under varying generation lengths, as shown in Table X. For InternVL2-8B, *SEC* reduces per-token latency by about 2 ms compared to standard autoregressive decoding. It also outperforms standard SPD across the board. The improvement is even more significant for the larger InternVL2-26B model while maintaining high generation accuracy. The significant acceleration stems from the tree-based parallel decoding, which reduces wasted verification computation and minimizes rollbacks.

To assess whether these efficiency gains come at the cost of semantic fidelity, we evaluate *SEC* on three widely adopted multimodal benchmarks: MME [83], MMVet [84], and MMMU [85]. As shown in Table XI, these results confirm that *SEC* is not limited to video analysis but also serves as a powerful accelerator for general image-text reasoning.

G. Ablation Studies

To gain deeper insights into components of our proposed *SEC* framework, we conduct a series of ablation studies

TABLE XI

COMPARISON RESULTS OF INTERNVL2-2B AND 26B ON MME, MMVET, AND MMMU

Methods	Type	Image Benchmark		
		MME	MMVet	MMMU
$\mathcal{M}_o$ InternVL2-8B [19]	Vanilla	2210.3	60.1	51.8
$\mathcal{M}_d$ InternVL2-2B [19]	<i>SEC</i>	2193.8	59.9	51.6
$\mathcal{M}_o$ InternVL2-26B [19]	Vanilla	2258.4	64.1	51.5
$\mathcal{M}_d$ InternVL2-2B [19]	<i>SEC</i>	2231.6	63.8	51.2

TABLE XII

ABLATION STUDY ON COMPONENTS OF *SEC*. WE EVALUATE THE IMPACT OF EDGE-SIDE FRAME SELECTION AND CLOUD-SIDE ADAPTIVE DECODING INDIVIDUALLY

Methods	Video Benchmark		Latency (ms/token)
	VideoMME	LVBench	
Vanilla Inference	55.4	54.3	108.7
+ Edge Frame Selection	57.7	57.9	108.9
+ Cloud Adaptive Decoding	57.6	57.7	74.1

TABLE XIII

ABLATION OF EDGE FRAME SELECTION COMPONENTS. WE EVALUATE THE IMPACT OF EACH STAGE IN OUR SEMANTIC-AWARE FRAME SELECTION PIPELINE: STARTING WITH UNIFORM SAMPLING AS THE BASELINE, FOLLOWED BY FRAME-QUERY MATCHING TO ALIGN FRAMES WITH USER QUERIES, AND FINALLY COMPUTING MARGINAL RELEVANCE TO ENHANCE TEMPORAL AND VISUAL DIVERSITY

Methods	Video Benchmark		Latency (ms/token)
	VideoMME	LVBench	
Uniform Sampling	55.4	54.3	74.0
+ Frame-Query Matching	56.9	56.1	74.1
+ Marginal Relevance Computing	57.6	57.7	74.1

on both the edge-side frame selection and cloud-side SPD modules, using InternVL2-26B with 16 frames on VideoMME and LongVideoBench with InternVL2-2B as the Draft Model. The per-token latency is measured over the generation of 256 output tokens.

1) *Main Components Ablations*: We evaluate the two core components of *SEC*: 1) semantic-aware edge frame selection and 2) adaptive cloud-side SPD. As shown in Table XII, adding only the edge selection module to vanilla inference improves accuracy by 2.3% on VideoMME and 3.6% on LongVideoBench, indicating that removing redundant frames substantially enhances MLLM input quality. This gain incurs only a minor latency increase, from 108.7 to 108.9 ms/token, due to lightweight frame-query scoring on the edge. With adaptive SPD enabled, latency drops sharply from 108.9 to 74.1 ms/token, while accuracy remains essentially unchanged (57.6% versus 57.7%). This confirms that adaptive speculation accelerates inference without harming reasoning fidelity. Overall, the edge and cloud modules are complementary; the edge module ensures high-quality inputs, while the cloud module delivers fast and efficient decoding.

2) *Edge Frame Selection Ablation*: Next, we investigate the contribution of each stage within the edge-side frame

TABLE XIV
ABLATION OF EDGE SELECTION CANDIDATES

Methods	Short-horizon		Long-horizon
	VideoMME-CP	PerTest-MC	LongVideoBench
Random Frame	47.0	63.1	57.3
First Frame	47.1	63.3	57.5
Center+3 Frame	47.1	63.2	57.6
Center-3 Frame	47.2	63.3	57.7
Last Frame	47.1	63.4	57.5
Center Frame	47.1	63.3	57.7

TABLE XV

ABLATION STUDY OF CLOUD-SIDE ADAPTIVE SPD. WE EVALUATE THE CONTRIBUTION OF EACH MODULE IN OUR ADAPTIVE SPD FRAMEWORK ON THE CLOUD, BUILT UPON A VANILLA MLLM WITH EDGE-SELECTED FRAMES. SEVERAL VARIANTS ARE EVALUATED, INCLUDING AUTO-REGRESSION, CONFIDENCE ROUTING, ATTENTION SHIFT ROUTING, AND TREE-BASED DRAFT AND VERIFICATION

Methods	Video Benchmark		Latency (ms/token)
	VideoMME	LVBench	
Auto-Regression	57.7	57.9	108.9
+ Confidence Routing	57.2	57.1	92.0
+ Attention Shift Routing	57.4	57.2	88.7
+ Tree-based Draft Verification	57.6	57.7	78.4
+ Tree-based Draft Generation	57.6	57.7	74.1

selection pipeline, as detailed in Table XIII. We start from uniform sampling as the baseline, where frames are selected at fixed intervals regardless of content. Adding frame-query matching increases accuracy by 1.5% on VideoMME and 1.8% on LVBench, showing that aligning frames with the user query improves task relevance.

Further integrating marginal relevance computing brings an additional 0.7% gain in accuracy, highlighting the importance of temporal and visual diversity. Without this step, the selected frames may cluster around a single event, missing other critical moments in the video. Moreover, the latency increases only marginally, from 74.0 to 74.1 ms/token, indicating that the computational overhead of the full selection process is minimal and acceptable for real-time deployment. The results confirm that our semantic-aware selection strategy is essential for achieving both high accuracy and efficient edge processing.

3) *Cloud-Side SPD Ablation*: We dissect the cloud-side SPD framework to assess the impact of its key components, as summarized in Table XIV. We conduct a fine-grained ablation study starting from the vanilla autoregressive inference baseline on the cloud, which achieves 57.7% accuracy on VideoMME with a latency of 108.9 ms/token.

When integrating confidence-based routing, the system dynamically avoids verification during high-confidence generation phases, reducing latency to 92.0 ms/token while maintaining competitive accuracy of 57.2%. This demonstrates that static verification intervals can be replaced with content-aware decisions to improve efficiency without sacrificing reliability. Further incorporating attention shift detection enables the model to proactively identify semantic transitions and trigger verification only when necessary, bringing latency

TABLE XVI
ABLATION STUDY ON CANDIDATE SEGMENT NUMBER M . WE USE $M = 64$ AS THE DEFAULT SETTING

Segment Num.	Video Benchmark		Edge Latency (ms)
	VideoMME	LVBench	
32	56.9	57.2	8.9
64	57.6	57.7	9.4
128	57.6	57.8	12.5
256	57.8	57.7	16.8

TABLE XVII

ABLATION STUDY ON TRADEOFF FACTORS α AND γ . WE SET BOTH FACTORS TO 0.8 BY DEFAULT

α	VideoMME	LVBench	γ	VideoMME	LVBench
0.2	56.9	57.1	0.6	56.8	57.0
0.4	57.2	57.4	0.7	57.4	57.4
0.6	57.4	57.4	0.8	57.6	57.7
0.8	57.6	57.7	0.9	57.1	57.5
1.0	57.3	57.2	1.0	57.1	57.3

down to 88.7 ms/token. The combination of both routing signals ensures that verification is performed neither too early nor too late, striking an optimal balance between speed and correctness.

Building upon this adaptive routing mechanism, the tree-based parallel SPD drafts and validates in parallel through a tree-structured computation pipeline, further lowering latency from 88.7 to 74.1 ms/token. During the draft, the draft model generates diverse candidate branches in a tree structure, enabling more opportunities for long-sequence acceptance. During validation, the target model validates multiple candidate sequences in parallel through a tree-structured KV-cache, significantly reducing redundant computation.

4) *Ablation on Frame Selection Candidates*: As shown in Table XV, we analyze the effect of the candidate number M and the candidate-frame construction in frame selection. Increasing M enlarges the semantic candidate pool and improves accuracy, but also increases edge-side latency. Accuracy improves noticeably when M increases from 32 to 64, while further enlarging M yields marginal gains with a significant latency increase; thus, we set $M = 64$ by default.

We also investigate different candidate generation strategies. As shown in Table XVI, simple extensions such as center ± 3 frames, as well as first/last/random frames, lead to very similar performance on both short-horizon (VideoMME-CP and PerTest-MC) and long-horizon (LongVideoBench) benchmarks (differences within $\sim 0.4\%$). This indicates that within short temporal intervals, frames are highly similar, and the choice of specific frames has little impact on performance.

5) *Ablation on Frame Selection Tradeoff Factors*: Table XVII studies the impact of the tradeoff coefficients α and γ . We observe that performance consistently improves as α increases, reaching its optimum at $\alpha = 0.8$, indicating the importance of temporal diversity in marginal relevance. Similarly, the best performance is achieved at $\gamma = 0.8$, suggesting that a balanced combination of semantic relevance

TABLE XVIII

ABLATION STUDY ON MAX DEPTH OF THE DRAFT TREE. WE USE $L = 6$ AS THE DEFAULT SETTING

Max Depth	VideoMME	LVBench	ACC. Rat.	Latency (ms/token)
1	50.9	51.1	56.4	51.9
2	52.6	53.4	58.2	59.1
4	56.1	55.9	61.9	69.4
6	57.6	57.7	62.7	74.1
8	57.7	57.7	65.8	87.9
10	57.7	57.9	68.4	117.4
12	57.7	57.9	70.2	179.0

TABLE XIX

ABLATION STUDY ON THE BRANCH NUMBER OF THE DRAFT NODE. WE USE $q = 3$ AS THE DEFAULT SETTING

branch Num.	VideoMME	LVBench	ACC. Rat.	Latency (ms/token)
1	56.2	56.9	50.2	99.7
2	56.9	57.3	53.7	84.6
3	57.6	57.7	62.7	74.1
4	57.6	57.9	64.8	94.9
5	57.7	57.9	69.2	146.7

and diversity-aware ranking is critical. Accordingly, we set both α and γ to 0.8 in all experiments.

6) *Ablation on Draft Tree of SPD*: To understand how the depth of the draft tree influences the effectiveness of SPD, we vary the maximum depth L and report results in Table XVIII. Increasing the depth allows the system to explore longer speculative trajectories before verification, which in turn improves its ability to recover from potential mismatches. As shown in the table, too-shallow trees significantly weaken performance because the model cannot backtrack to earlier branching points, leading to suboptimal acceptance ratios and inferior accuracy on both VideoMME and LongVideoBench. As the depth grows, accuracy steadily improves and reaches near-saturation around $L = 6$, achieving 57.6% on VideoMME and 57.7% on LongVideoBench. However, deeper trees bring accuracy gains while causing a substantial increase in latency due to the exponential growth of unverified speculative branches. Balancing performance improvements with overhead, we select $L = 6$ as the default configuration.

We further examine the effect of the number of speculative candidates expanded at each node of the draft tree. As summarized in Table XIX, small values of q (e.g., $q = 1, 2$) constrain hypothesis diversity, reducing the probability that the ground-truth continuation is covered by the draft and consequently yielding lower acceptance ratios and slight accuracy degradation. Performance peaks at $q = 3$, where the model achieves an optimal tradeoff between coverage and verification overhead. It attains the accuracy of 57.6% on VideoMME and 57.7% on LongVideoBench, and the acceptance ratio of 62.7%. Beyond this point, increasing q to 4 or 5 introduces diminishing returns, while end-to-end latency rises sharply (e.g., + 72.6 ms/token from $q = 3$ to $q = 5$, reaching 146.5 ms/token), primarily due to increased verification and pruning costs. Given this efficiency–accuracy balance, we adopt $q = 3$ as the default configuration in all main experiments.

VII. LIMITATION AND FUTURE WORK

Although *SEC* significantly improves efficiency in edge–cloud collaborative video understanding, it relies on semantic-aware keyframe selection, which may miss critical visual events. In particular, rare or short-lived events, small objects, or tasks requiring high temporal precision may not be fully captured by relevance-driven sampling. Addressing this limitation remains an important direction for future work.

Beyond this limitation, several promising research directions remain open.

- 1) *Sensitive Data Protection and Privacy-Preserving IoT Analytics*: As IoT systems increasingly incorporate personal, medical, and behavior-sensitive information, ensuring secure data transmission and privacy protection becomes a critical requirement [86], [87], [88]. Future extensions of *SEC* could integrate privacy-preserving mechanisms to provide verifiable and tamper-resistant edge–cloud data pipelines.
- 2) *Graph Neural Networks (GNNs) for Cross-Modal and Multisource Fusion*: The current *SEC* focuses on optimizing frame selection and decoding, but does not explicitly model relational structures across heterogeneous data sources [88], [89]. Integrating GNNs into *SEC* could enable richer cross-modal semantic fusion and improved robustness under missing or sparse data scenarios.

VIII. CONCLUSION

In this article, we propose *SEC*, a semantic-aware edge–cloud collaborative framework for low-latency intelligent video analysis in resource-constrained IoT environments. By addressing the dual bottlenecks of excessive data transmission at the edge and high inference latency in cloud-based MLLMs, *SEC* achieves efficient and accurate video understanding through collaborative computation. On the edge side, our lightweight semantic-aware frame selection module identifies task-relevant key frames, effectively filtering out redundant or irrelevant content while preserving essential information. On the cloud side, we proposed an adaptive SPD framework with tree-based parallel decoding, accelerating the decoding rates. Extensive experiments on VideoMME and LongVideoBench under realistic edge–cloud deployment settings demonstrate that *SEC* significantly reduces the end-to-end latency while improving the accuracy. Notably, *SEC* operates in a training-free manner and is easily adaptable to arbitrary MLLMs, offering strong practicality and plug-and-play compatibility for real-world deployment. Our work bridges the gap between high-performance multimodal understanding and real-time system requirements, paving the way for scalable, responsive, and intelligent video analytics in future IoT ecosystems.

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